

From Gates To FPGA, Deeds Teaches COMPLETE DIGITAL DESIGN

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Logic design is a vital part of modern electronics curriculum. Through this article, we try to familiarise our readers with a smart digital system design environment called Deeds, or Digital Electronics Education and Design Suite.

Deeds includes a set of simulators, tools and learning guides to help academia get a solid understanding of the principles of digital design. The software was developed at Department of Electrical, Electronic, Telecommunications Engineering and Naval Architecture (DITEN), University of Genoa, Italy. The latest release of the design suite version 1.95.500 is included in the DVD accompanying this month's EFY Plus.

Simulators used

The three simulators that could be found in Deeds are Deeds-DcS (digital-circuit simulator), Deeds-FsM (finite state machine) and Deeds-McE (microcomputer emulator). With the help of Deeds, a user can easily design complex networks with a combination of standard logic, state machines and embedded microcomputers.

Deeds-DcS is a user-friendly circuit simulator with a graphical schematic editor and a comprehensive library of logic components. As the purpose

of the design suite is to cater to the needs of academia, inclusion of commercial part Deeds-DcS library is based on standard logic functions. Within the library, there can be user-definable devices implemented as finite-state machines or 8-bit microcomputer components for embedded and interrupt based systems.

Finite-state machine components defined by Deeds-FsM tool as per

algorithmic-state machine (ASM) paradigm can be saved in files and could then be directly imported as the component of schematic editor in Deeds-DcS.

Users can find all necessary components for an Assembly language based microcomputer programming inside Deeds-McE. They can find a source code editor with syntax highlighting, assembler and interactive debugger with Assembly code in execution, central processing unit registers, memory contents and I/O ports state. The debugger can run programs all at once or step by step. A single line of code could be used to assemble, link and load the program in the emulated system memory.

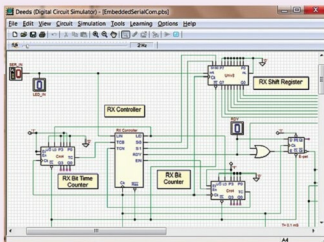


Fig. 1: A digital system designed with the help of Deeds-FsM

A quick glance at Deeds

Software	: Digital Electronics Education and Design Suite
Type	: Digital electronics learning environment
Areas covered	: Combinational logic networks Sequential logic networks Finite-state machine design Microcomputer programming (at Assembly level) Microcomputer interfacing FPGA programming
Main developers	: Giuliano Donzellini and team
Developed at	: DITEN, University of Genoa, Italy
Latest stable release	: Version 1.95.500 (Jan 02, 2016)
Website	: www.esng.dibe.unige.it/deeds

The latest version

Users can find a whole new set of digital-to-analogue-converter (DAC) components in the latest version of Deeds. Adders, counters, timers, random access memory (RAM), read only memory (ROM) and other enhanced components are also included in the library. Let us take a quick look at some interesting inclusions.

DAC. DAC component is symbolised by a pentagon on the left side and a square empty area at the right. At the time of simulation, converted values